

IoT Architecture for Smart Agriculture: Modeling, Communication and Data Analytics

A. Makhoul
Department of Computer Science

June 19, 2026

Abstract

The Internet of Things is gradually transforming agricultural infrastructures into intelligent systems capable of measuring, communicating and making decisions in real time. This article presents a complete architecture for connected agriculture, from the perception layer to cloud-based data analytics. We describe communication protocols adapted to constrained devices, propose a simplified mathematical model of soil moisture, and illustrate the use of learning algorithms to predict irrigation requirements. The document deliberately contains many typographical elements useful for a reconstruction exercise in $\text{\LaTeX}$: sections, subsections, figures, tables, equations, algorithms, cross-references and bibliography.

Keywords: Internet of Things, smart agriculture, MQTT, LoRaWAN, sensors, machine learning, security.

Contents

1	Introduction	2
2	General Architecture of an IoT System	2
2.1	Perception Layer	2
2.2	Network Layer	2
2.3	Application Layer	3
3	Communication Protocols	3
3.1	MQTT	3
3.2	CoAP and HTTP	3
3.3	Comparative Analysis	3
4	Mathematical Modeling of Soil Moisture	4
4.1	Numerical Discretization	4
4.2	System Stability	4
5	Machine Learning for Prediction	4
5.1	Linear Regression	4
5.2	Neural Networks	4
6	Data Collection and Decision Algorithms	5
7	Experimental Results	5
7.1	Setup	5
7.2	Model Comparison	5
8	Security and Privacy	6
8.1	Main Threats	6
8.2	Risk Assessment	6
9	Discussion	7
10	Conclusion	7
A	Additional Experimental Data	7

List of Figures

1	Simplified architecture of an IoT system for smart agriculture.	3
2	Examples of visualizations produced from sensor data.	5
3	Prediction error for three learning models.	6

List of Tables

1	Comparison of protocols used in IoT systems.	3
2	Characteristics of the prototype sensors.	6
3	Comparison of moisture prediction models.	6
4	Extract of data collected by the sensors.	7

1 Introduction

The Internet of Things refers to a set of physical devices equipped with sensors, microcontrollers and communication modules capable of exchanging data with gateways or remote platforms. In agriculture, this technology makes it possible to monitor soil moisture, temperature, light intensity, plant growth and water consumption. The generated measurements are then used to optimize irrigation, reduce losses and improve productivity.

The objective of this article is twofold. On the one hand, it provides a scientific overview of the design of an IoT system for smart agriculture. On the other hand, it offers a complete support document for a practical L^AT_EX assignment. Students can therefore reconstruct an article containing figures, equations, tables, cross-references, algorithms and a bibliography.

The rest of the article is organized as follows. Section 2 presents the general architecture. Section 3 compares several communication protocols. Section 4 introduces a mathematical model of soil moisture. Section 6 gives a data collection and decision algorithm. Finally, Sections 7 and 8 discuss the experimental results and security issues, respectively.

2 General Architecture of an IoT System

An IoT architecture is often structured into three main layers: the perception layer, the network layer and the application layer. This decomposition facilitates system design, performance analysis and the identification of security risks.

2.1 Perception Layer

The perception layer includes the sensors and actuators installed in the field. Sensors measure physical quantities such as temperature, air humidity, soil moisture, solar radiation or water level. Actuators control solenoid valves, pumps or ventilation systems.

Definition 1 (Sensor node). *A sensor node is an embedded device composed at least of a sensor, a microcontroller, an energy source and a wireless communication interface.*

On a farm, sensor nodes must be robust, inexpensive and energy autonomous. Electrical consumption is therefore a central constraint, especially when the devices are powered by batteries or solar panels.

2.2 Network Layer

The network layer transports data from sensors to a gateway or a remote platform. The choice of protocol depends on range, throughput, energy consumption and deployment cost. In rural areas, LoRaWAN is often relevant for transmitting small amounts of data over several kilometers. For greenhouses or agricultural buildings, Wi-Fi or ZigBee may be sufficient.

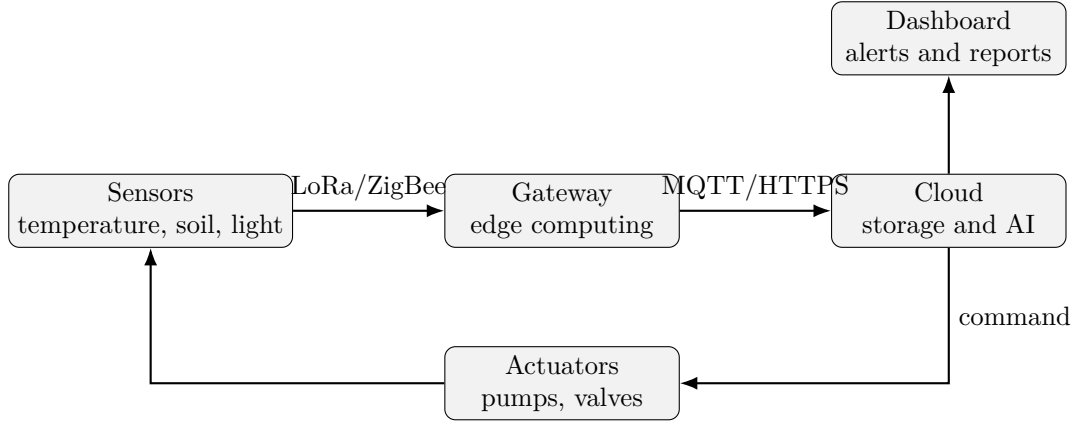


Figure 1: Simplified architecture of an IoT system for smart agriculture.

2.3 Application Layer

The application layer provides visualization, storage and decision-support services. It includes dashboards, time-series databases, predictive models and automation rules. For example, a simple rule can trigger irrigation when soil moisture falls below a given threshold.

Figure 1 illustrates a complete cycle: collection, transmission, analysis, decision and action. In an industrial system, this cycle must remain reliable even when connectivity is intermittent.

3 Communication Protocols

Connected objects generally have limited resources. It is therefore necessary to use lightweight protocols adapted to constrained environments. This section compares MQTT, CoAP, HTTP and LoRaWAN.

3.1 MQTT

MQTT is a protocol based on the publish/subscribe model. Sensors publish messages to topics, while client applications subscribe to topics of interest. A server, called a *broker*, receives and redistributes the messages.

The efficiency of a transmission can be described by the ratio

$$\eta = \frac{P_d}{P_d + P_o}, \quad (1)$$

where P_d denotes the useful data size and P_o the overhead introduced by protocol headers. When P_o increases, efficiency decreases, which can become problematic for low-throughput networks.

3.2 CoAP and HTTP

CoAP is designed for constrained devices and is based on UDP. It provides an interface close to the REST model, but with a lower overhead than HTTP. HTTP remains relevant when objects are not highly constrained or when interoperability with conventional web services is a priority.

3.3 Comparative Analysis

Table 1 compares four common technologies. It shows that the optimal protocol strongly depends on the usage context.

Table 1: Comparison of protocols used in IoT systems.

Protocol	Transport	Range	Consumption	Typical Use
MQTT	TCP	medium	low	cloud telemetry
CoAP	UDP	medium	very low	constrained objects
HTTP	TCP	medium	high	web services
LoRaWAN	LPWAN radio	long	very low	agricultural fields

4 Mathematical Modeling of Soil Moisture

To control irrigation, field measurements must be linked to a dynamic model. Let $M(t)$ denote soil moisture at time t . A simplified model can be written as

$$\frac{dM(t)}{dt} = I(t) - E(t) - D(t), \quad (2)$$

where $I(t)$ represents the water input due to irrigation, $E(t)$ evaporation and $D(t)$ drainage. This model is intentionally simple, but it highlights the main variables controlling moisture evolution.

4.1 Numerical Discretization

Using a time step h , an explicit Euler approximation of Equation (2) gives

$$M_{n+1} = M_n + h(I_n - E_n - D_n). \quad (3)$$

The mean squared error between a measurement y_i and a prediction $\hat{y}_i$ is defined by

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (4)$$

4.2 System Stability

Consider the linearized model

$$M_{n+1} = aM_n + bU_n, \quad (5)$$

where U_n denotes the irrigation control. Stability depends on the value of the coefficient a .

Proposition 1. *If $|a| < 1$ and if the control U_n is bounded, then the sequence (M_n) remains bounded.*

Proof. By iterating the recurrence relation, we obtain

$$M_n = a^n M_0 + b \sum_{k=0}^{n-1} a^{n-1-k} U_k.$$

If $|a| < 1$, the term $a^n M_0$ converges to zero. If $|U_k| \leq C$, the geometric sum is bounded by $C/(1 - |a|)$. Therefore, the sequence is bounded. $\square$

5 Machine Learning for Prediction

Machine learning models exploit historical measurements to predict future needs. Input variables may include temperature, air humidity, soil moisture, wind speed and weather forecasts.

5.1 Linear Regression

The linear regression model is written as

$$\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j, \quad (6)$$

where x_j are explanatory variables and β_j are model parameters. The parameters are usually estimated by minimizing the cost function

$$J(\beta) = \frac{1}{2N} \sum_{i=1}^N (y_i - \hat{y}_i)^2. \quad (7)$$

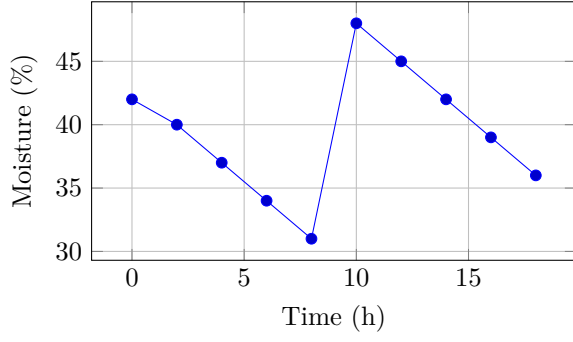
5.2 Neural Networks

An artificial neuron computes a weighted combination of the inputs, followed by an activation function:

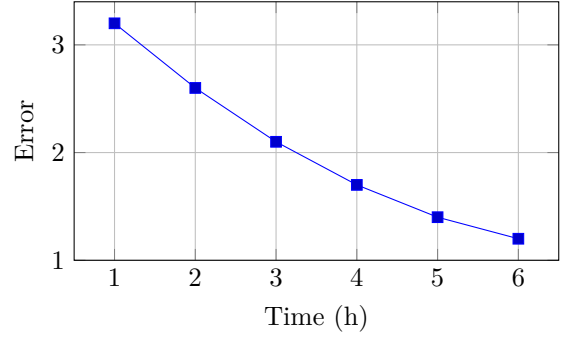
$$z = f \left(\sum_{j=1}^p w_j x_j + b \right). \quad (8)$$

Neural networks can learn nonlinear relationships, but they require more data and a more costly training phase.

Figure 2 combines two subfigures: the first represents a moisture time series and the second illustrates the decrease of an error during training.



(a) Raw measurements.



(b) Learning error.

Figure 2: Examples of visualizations produced from sensor data.

6 Data Collection and Decision Algorithms

The operation of an intelligent irrigation system can be summarized by a collection, transmission and decision algorithm. Algorithm 1 shows a simplified version.

Algorithm 1 Data Collection and Irrigation Decision

Require: Moisture threshold θ , sampling period Δt

Ensure: Irrigation decision and data recording

```

1: Initialize the sensors and the network connection
2: while the system is active do
3:   Read temperature  $T$ , air humidity  $H$  and soil moisture  $M$ 
4:   Transmit the triplet  $(T, H, M)$  to the gateway
5:   Predict future moisture  $\hat{M}$  using the selected model
6:   if  $\hat{M} < \theta$  then
7:     Activate the pump for a predefined duration
8:   else
9:     Keep the system in standby mode
10:  end if
11:  Wait for  $\Delta t$ 
12: end while

```

The time complexity of this algorithm is linear with respect to the number of measurement cycles. For N cycles, we obtain

$$T(N) = \sum_{i=1}^N c = cN = \mathcal{O}(N), \quad (9)$$

where c represents the average cost of one cycle.

7 Experimental Results

This section presents a fictitious but realistic set of results. The objective is to illustrate how experimental data can be presented in a scientific article. The measurements correspond to a plot equipped with moisture sensors and an automatically controlled pump.

7.1 Setup

The prototype includes six sensor nodes, a LoRa gateway and a storage platform. Measurements are sent every ten minutes. Table 2 describes the main sensors used.

7.2 Model Comparison

Table 3 compares three predictive models. The results show that the neural network obtains the lowest error, but at the cost of a higher training time.

The results in Figure 3 confirm the values in Table 3. In a real application, the choice of the model must take into account accuracy, energy cost and the computing capacity available on the gateway.

Table 2: Characteristics of the prototype sensors.

Sensor	Quantity	Range	Accuracy
DHT22	temperature	-40°C to 80°C	$\pm 0.5^{\circ}\text{C}$
DHT22	air humidity	0 to 100 %	$\pm 2\%$
SEN0193	soil moisture	0 to 100 %	$\pm 3\%$
BH1750	light intensity	1 to 65 535 lux	$\pm 20\%$

Table 3: Comparison of moisture prediction models.

Model	RMSE	MAE	Training Time
Linear regression	3.21	2.74	0.4 s
Random forest	1.87	1.45	3.1 s
Neural network	1.32	1.01	12.7 s

8 Security and Privacy

IoT systems are exposed to many attacks: traffic eavesdropping, identity spoofing, measurement modification, denial-of-service attacks and physical compromise of nodes. The consequences can be significant, because a wrong irrigation decision may damage a crop or waste a large amount of water.

8.1 Main Threats

Threats can be classified according to the targeted layer:

- **Perception:** measurement falsification, sensor replacement, battery exhaustion.
- **Network:** interception, message replay, radio jamming.
- **Application:** unauthorized access, data leakage, cloud misconfiguration.

8.2 Risk Assessment

If p is the probability of compromising one node and if the network contains n independent nodes, the probability that at least one node is compromised is

$$P(A) = 1 - (1 - p)^n. \quad (10)$$

This equation shows that the global risk increases with the size of the deployment. It is therefore necessary to provide authentication, encryption and software update mechanisms.

Theorem 1 (Risk reduction through authentication). *If an authentication mechanism divides the probability of compromising one node by a factor $\alpha > 1$, then the global risk defined in Equation (10) strictly decreases for all $n \geq 1$.*

Proof. After authentication, the individual probability becomes p/α . The new risk is $P'(A) = 1 - (1 - p/\alpha)^n$. Since $0 < p/\alpha < p < 1$, we have $(1 - p/\alpha)^n > (1 - p)^n$, and therefore $P'(A) < P(A)$. $\square$

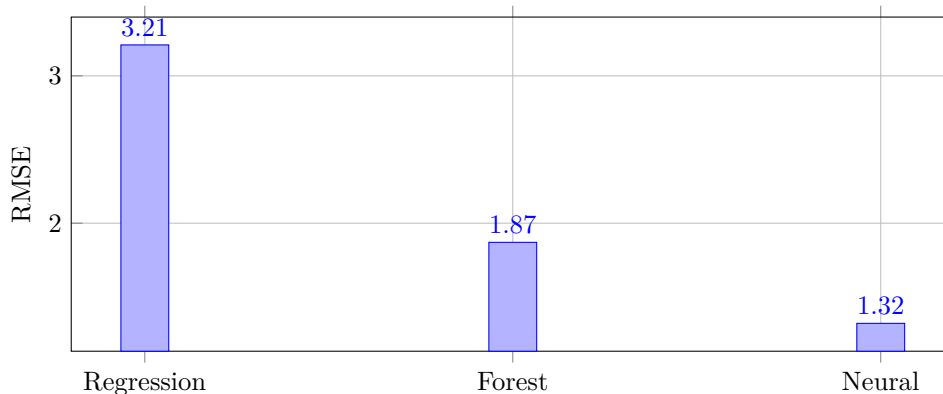


Figure 3: Prediction error for three learning models.

9 Discussion

Agricultural IoT architectures must reconcile three sometimes contradictory objectives: measurement accuracy, energy autonomy and communication reliability. A high sampling frequency improves data quality, but increases energy consumption. Similarly, a complex predictive model may improve the decision, but require more computation.

Several practical trade-offs may be adopted:

1. perform local preprocessing on the gateway;
2. dynamically adapt the measurement frequency;
3. use a long-range protocol for isolated areas;
4. apply security mechanisms from the design phase.

Data quality also remains a critical issue. Poorly calibrated or badly positioned sensors can introduce bias. It is therefore recommended to include a calibration phase, followed by automatic outlier detection.

10 Conclusion

This article has presented a complete IoT architecture for smart agriculture. We described the system layers, compared several communication protocols, proposed a mathematical model of soil moisture and illustrated the use of machine learning models. The fictitious experimental results show how to present tables, figures and metrics in a scientific article.

From a pedagogical point of view, this document covers the main L^AT_EX concepts needed for scientific writing: page layout, table of contents, lists of figures and tables, equations, mathematical environments, algorithms, cross-references, professional tables, subfigures, bibliography and appendices.

A Additional Experimental Data

Table 4 gives an example of a simplified measurement series. It can be used as a basis for reproducing the figures in the article.

Table 4: Extract of data collected by the sensors.

Time (h)	Temperature	Air Humidity	Soil Moisture	Irrigation
0	21.4	62	42	no
2	22.1	60	40	no
4	23.5	58	37	no
6	25.0	55	34	no
8	26.8	51	31	yes
10	25.9	53	48	no
12	24.7	57	45	no
14	23.8	59	42	no

B Citation and Cross-Reference Example

The MQTT protocol is often recommended for lightweight telemetry [3]. IoT architectures are generally studied through several layers, as indicated by Atzori and colleagues [1]. In this document, Figure 1, Table 1 and Equation (4) illustrate the use of cross-references.

References

- [1] L. Atzori, A. Iera and G. Morabito, “The Internet of Things: A survey”, *Computer Networks*, vol. 54, no. 15, pp. 2787–2805, 2010.
- [2] J. Gubbi, R. Buyya, S. Marusic and M. Palaniswami, “Internet of Things (IoT): A vision, architectural elements, and future directions”, *Future Generation Computer Systems*, vol. 29, no. 7, pp. 1645–1660, 2013.

- [3] A. Banks and R. Gupta, “MQTT Version 3.1.1”, OASIS Standard, 2014.
- [4] A. Zanella, N. Bui, A. Castellani, L. Vangelista and M. Zorzi, “Internet of Things for Smart Cities”, *IEEE Internet of Things Journal*, vol. 1, no. 1, pp. 22–32, 2014.
- [5] S. Wolfert, L. Ge, C. Verdouw and M.-J. Bogaardt, “Big Data in Smart Farming – A review”, *Agricultural Systems*, vol. 153, pp. 69–80, 2017.